Thinking for Theorizing:
How the Definitional Copular Frame Channels Processual Concepts into Entity-Search Reasoning

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Abstract

This paper introduces *Thinking for Theorizing*: the claim that grammatical structure constrains not everyday cognition as such, but the formal construction of theoretical questions. We examine the definitional copular frame "What is X?" — the canonical question form of much Western philosophical inquiry — as a grammatical funnel that channels processual concepts into nominalized, entity-like forms. Two studies provide converging evidence. In Study 1, a cross-linguistic analysis of 15 Universal Dependencies treebanks (~3.2 million tokens, five language families) cross-referenced with WALS copula typology data, we find that languages with shared nominal-locational predication and high copular versatility exhibit substantially higher rates of noun-subject copular constructions used as a structural proxy for the definitional frame (HIGH/LOW constraint ratio = 2.63×, large effect sizes for predicted comparisons). The effect is structural rather than genealogical: Indo-European versus non-Indo-European languages show no difference in copula rate (ratio = 1.00). In Study 2, a distributional-semantic analysis using hypothesis-free stimuli and two independent embedding models shows that invisible-process concepts such as *love*, *consciousness*, and *truth* occupy more entity-oriented — or less process-anchored — distributional positions than visible-process concepts such as *running* and *swimming* across all eight grammatical forms tested (8/8 cross-model agreement). The visible/invisible classification is independently validated against Brysbaert et al.'s (2014) concreteness ratings, and the effect is especially pronounced in definitional copular frames. English provides four classes of grammatical escape routes from reification — processual (-ing), agentive (-er), gradable adjectival (-ful, -able, -ous), and infinitival (to-V) — but many canonical formulations in Western philosophy privilege the bare-noun copular form. We argue that this convergence reflects not grammatical determinism but a co-evolutionary account in which grammatical affordances, the institutional objective of definition, the social requirements of communal discourse, and the observational invisibility of certain processes jointly stabilize entity-search reasoning as the default mode of theoretical inquiry into invisible concepts.

Keywords: definitional copular frame; nominalization;

linguistic relativity; embedding analysis; conceptual reification; Thinking for Theorizing; cross-linguistic typology

1. Introduction

1.1 A recurring question form

Consider a small sample of questions drawn from the canonical literature of Western philosophy:

- *What is justice?* (Plato, *Republic*)
- *What is beauty?* (Plato, *Hippias Major*; Hume, "Of the Standard of Taste")
- *What is virtue?* (Plato, *Meno*)
- *What is love?* (Plato, *Symposium*)
- *What is knowledge?* (Plato, *Theaetetus*; Gettier, 1963)
- *What is truth?* (Aquinas, *Summa Theologiae*; Tarski, 1944)
- *What is the self?* (Locke, *Essay Concerning Human Understanding*; Hume, *Treatise*)
- *What is being?* (Aristotle, *Metaphysics*; Heidegger, *Sein und Zeit*)
- *What is freedom?* (Kant, *Critique of Practical Reason*; Berlin, 1958)
- *What is meaning?* (Frege, "Über Sinn und Bedeutung"; Wittgenstein, *Philosophical Investigations*)
- *What is consciousness?* (Chalmers, 1995)

These questions span twenty-four centuries and a wide range of philosophical traditions. They differ in subject matter, methodology, and the conceptual schemes they presuppose. But they share an immediately visible structural property: each is constructed in the form **"What is X?"**, where X is a bare noun naming a concept whose referent is not directly observable.

This structural uniformity is not a coincidence. It is, we argue, the surface manifestation of a deeper convergence — one in which grammatical structure, institutional practice, and the invisibility of the concepts under inquiry jointly select a particular question form. The same pattern is conspicuously absent from questions about observable processes. Philosophers have not, in any institutionally significant way, asked *What is running?*, *What is eating?*, or *What is dancing?* These activities, no less ubiquitous in human life, do not generate the same form of question — and, as we shall argue, this asymmetry has a structural explanation.

The questions in the list above share a further property: their referents are, in everyday usage, predominantly *processual.* One *acts justly*, *finds something beautiful*, *behaves virtuously*, *loves another*, *comes to know*, *tells the truth*, *recognizes oneself*, *exists*, *becomes free*, *means something by an utterance*, *is conscious of* a situation. The nominalizations that transform these processes into *justice*, *beauty*, *virtue*, *love*, *knowledge*, *truth*, *self*, *being*, *freedom*, *meaning*, and *consciousness* are grammatical operations — and crucially, ones that the language in question does not require.

English, the language in which much of contemporary Anglophone philosophy is conducted, provides multiple processual alternatives. Progressive forms (*being just*, *loving*, *being conscious*) preserve temporal continuity. Agentive forms (*a lover*, *a knower*, *a thinker*) re-attach the process to a visible human agent. Gradable adjectival forms (*just*, *beautiful*, *truthful*, *meaningful*, *conscious*) preserve degree structure and continuous variability. Infinitives (*to love*, *to know*, *to be conscious*) maintain action-event structure. All of these alternatives are grammatically available. Yet many canonical formulations in

Western philosophy privilege a particular alternative: the bare-noun copular form, placed in the definitional frame "What is X?"

This paper investigates the hypothesis that the definitional copular frame functions as a **grammatical funnel**: a structural channel that channels processual concepts into nominal form, providing a semantic precondition for entity-search reasoning — that is, reasoning that seeks the essence, nature, or definition of a presumed entity. We provide converging evidence from cross-linguistic corpus analysis (Study 1) and distributional-semantic analysis (Study 2) that the funnel operates selectively: it shifts invisible-process concepts (those whose referents cannot be directly observed) from process-space into entity-space, while leaving visible-process concepts in process-space even when nominalized.

Our central claim is not that grammar *determines* philosophical thought. Rather, grammar provides a set of formal affordances, among which philosophical institutions have selectively stabilized the bare-noun copular form because it efficiently serves the institutional objective of definition. The recurring question form documented in the list above emerges, we argue, from a co-evolutionary loop in which grammatical availability, institutional demand for definition, social convenience of a unified question-subject, and the conceptual invisibility of the target jointly stabilize "What is X?" as the default form of theoretical inquiry into processual concepts.

1.2 Relation to existing theories

The claim that language shapes thought has a long history in linguistic and cognitive science, most prominently associated with the Sapir-Whorf hypothesis. The present investigation occupies adjacent but structurally distinct territory. We here distinguish the present account from three lines of existing work: classical lexical-level relativity claims, Slobin's "Thinking for Speaking," and Boroditsky's conceptual-metaphor research. These distinctions are summarized in Table 1 and elaborated in the subsections that follow.

Table 1. Distinction from existing linguistic relativity theories.

Theory	Locus of effect	Mechanism	Domain affected
Sapir-Whorf, lexical	Vocabulary	Word categories shape perceptual categorization	Color, space, number perception
Sapir-Whorf, grammatical	Obligatory grammatical marking	Required encoding shapes attention	Habitual categorization
Slobin (1996)	Syntax + lexicon	Grammar shapes online encoding for speech	Communication / speech production
Boroditsky and colleagues	Conceptual metaphor + grammatical gender	Mapping shapes offline reasoning	Temporal cognition, gender attribution
Present account	Definitional copular frame	Frame channels concepts into entity-space	Theoretical / philosophical question formulation

1.2.1 *Lexical-level relativity*

The classical Sapir-Whorf research program operated principally at the lexical level. Whorf's (1956) original arguments concerning Hopi tense, Berlin and Kay's (1969) cross-linguistic study of color terms, and the subsequent literature on color cognition (e.g., Kay & Kempton, 1984; Roberson et al., 2000) examined whether differences in vocabulary correlated with differences in perceptual discrimination or categorization. The general pattern of findings is that lexical distinctions do produce small but measurable effects on perceptual categorization, particularly at category boundaries (Winawer et al., 2007), but the effects are modest and often disappear under verbal interference paradigms (Gilbert et al., 2006).

Levinson and colleagues (Levinson, 2003; Pederson et al., 1998) extended this work to spatial cognition, demonstrating that speakers of languages using absolute spatial frames (north/south/east/west) versus relative frames (left/right) exhibit corresponding differences in nonlinguistic spatial reasoning. Lucy (1992, 1996) similarly demonstrated grammatical-number effects on categorization in Yucatec Maya vs. English speakers.

These programs share a structural feature: the locus of the cross-linguistic effect is the **lexicon** (or in Lucy's case, obligatory grammatical categories), and the **domain affected** is perceptual or categorization-level cognition. The proposed mechanism is some form of habituated attention or categorization shaped by the words a speaker frequently uses.

The present account operates at a different locus and targets a different domain. The locus is not the lexicon but the **syntactic construction** — specifically, the definitional copular frame "What is X?" The domain affected is not perceptual categorization but **the formulation of theoretical questions**. Speakers of high-constraint and low-constraint languages may perceive love, justice, or beauty in indistinguishable ways at the level of phenomenal experience; the difference, on the present account, lies in the structural prominence of the question form that institutional discourse uses to inquire about these phenomena.

This is also why our findings are not undermined by the broader critique of strong lexical relativity. The "great Eskimo vocabulary hoax" (Pullum, 1989) demonstrated that the lexical-count argument for Whorfian effects rested on inflated and inaccurate vocabulary counts. The present claim does not rest on vocabulary counts; it rests on the cross-linguistically variable prominence of a particular syntactic construction (Sections 3.2.1–3.2.4) and on the distributional-semantic consequences of nominalization specifically for invisible-process concepts (Study 2).

1.2.2 *Slobin's "Thinking for Speaking"*

Slobin (1996, 2003) proposed a reformulation of linguistic relativity that shifted attention from offline cognition (perception, memory, categorization) to **online speech production**. The "Thinking for Speaking" hypothesis holds that grammatical structure does not shape general cognition but does shape what speakers attend to and encode during the act of speaking. A Spanish speaker preparing to describe a motion event must specify path information that an English speaker may leave unspecified; conversely, an English speaker must specify manner information that a Spanish speaker may leave unspecified. The grammatical structure of each language thus shapes the *online* construction of utterances.

Slobin's account explicitly disavows the strong Whorfian claim that grammar shapes thought in general. The effects are confined to the moment of speech production. Subsequent extensions of the framework (Berman & Slobin, 1994; Slobin, 2003) examined cross-linguistic differences in narrative construction, demonstrating that speakers of typologically different languages produce systematically different narrative structures even when describing the same events.

The present account shares with Slobin's the move from lexicon to grammar as the locus of cross-linguistic effects. But the **domain** differs in three respects.

First, Slobin's domain is *online communication*: the immediate, time-pressured construction of utterances in real-time speech. The present domain is *offline theorization*: the deliberate, institutionally reinforced construction of theoretical questions through composition, revision, citation, and canonical transmission. These are very different processing contexts.

Second, Slobin's mechanism is **selection-for-encoding**: speakers select which features of an event to encode based on what their grammar makes obligatory or salient. The present mechanism is **frame-driven nominalization**: the definitional copular frame requires its target to be a nominal, and processual concepts are nominalized to fill this position. The mechanisms operate at different levels of linguistic structure.

Third, Slobin's empirical methodology examines speech production data — what speakers actually say when describing events. The present methodology examines corpus distributions and embedding-space positioning of grammatical forms, with the inferential target being not what speakers say but what becomes institutionally canonical in theoretical discourse.

We retain Slobin's framing — "Thinking for X" — but substitute *Theorizing* for *Speaking*. The change is not merely terminological. *Speaking* describes a real-time skilled performance under cognitive pressure. *Theorizing* describes an extended, deliberate, institutionally mediated practice. The latter is far more susceptible to selection by institutional pressures (Section 6.1), and far less susceptible to the cognitive constraints that shape online speech production.

1.2.3 Boroditsky and the metaphor program

Lera Boroditsky and colleagues have produced a substantial body of work examining whether grammatical structure shapes offline reasoning, particularly through the mediation of conceptual metaphor. Boroditsky (2001) demonstrated that Mandarin speakers, whose language often uses vertical metaphors for time (上 *shàng* for earlier, 下 *xià* for later), were faster than English speakers (whose language predominantly uses horizontal metaphors) to verify temporal sequences after vertical priming. Casasanto and Boroditsky (2008) extended this work to duration perception. Boroditsky et al. (2003) demonstrated grammatical-gender effects on object attribution: Spanish speakers, whose word for "key" is masculine, were more likely to describe keys with masculine-stereotypical adjectives than German speakers, for whom "key" is feminine.

This program shares with the present account a willingness to locate linguistic effects in offline, deliberate reasoning rather than in perception alone. It differs in mechanism. Boroditsky's mechanism operates through **conceptual metaphor**: a grammatical pattern (vertical vs. horizontal spatial expressions; gender assignment) primes a conceptual mapping (time as vertical/horizontal; objects as masculine/feminine), and the mapping shapes downstream reasoning. The mechanism is mediational: grammar → metaphorical mapping → reasoning.

The present mechanism is structurally different. The definitional copular frame does not metaphorically map a concept onto another domain. It **nominally positions** a concept — that is, it requires the concept to occupy the syntactic position of a noun, which then licenses the kinds of operations standardly applied to nouns (definition-seeking, essence-attribution, existence-questioning). The path is grammatical → distributional-semantic positioning → question-formulation, not grammatical → metaphorical mapping → reasoning.

A second difference concerns the **domain of inquiry**. Boroditsky's program has primarily examined everyday concepts (time, gender, motion). The present account targets a more restricted domain: the question forms used in theoretical and philosophical discourse. The two programs are complementary rather than competing. Boroditsky's results show that grammatical structure can shape offline reasoning about ordinary concepts; the present results show that grammatical structure correlates with the structural form of canonical theoretical questions.

A third difference concerns **scope of effect**. Boroditsky's findings are typically modest in magnitude — small but reliable differences in reaction time or rating. The present findings, by contrast, involve structural rather than gradational claims: certain question forms either are or are not canonical in a tradition. The scale of measurement is qualitatively different.

1.2.4 "Thinking for Theorizing"

Drawing the threads together, we introduce **"Thinking for Theorizing"** as a label for the present account. The phrase echoes Slobin's "Thinking for Speaking" but substitutes a different cognitive activity. *Theorizing*, as we use the term, refers to the deliberate, institutionally embedded, citation-bearing, canonical-transmission-oriented practice of constructing and answering questions about concepts whose referents are not immediately observable. This is the activity of philosophy, but it is also, in modified forms, the activity of theoretical disciplines in the sciences (theoretical physics, theoretical biology, theoretical economics) and the humanities (literary theory, social theory).

The hypothesis is that grammatical structure constrains not everyday cognition (which is flexible across forms and contexts) nor online communication (where Slobin's mechanisms operate) but the **formal structure of theoretical questions** as they emerge, stabilize, and propagate through institutional discourse. The mechanism is the grammatical funnel articulated in Section 2.1. The domain is the canonical literature of theoretical inquiry. The evidence, presented in Studies 1 and 2, is corpus-distributional and distributional-semantic.

This formulation positions the present account within the existing literature on linguistic relativity without inheriting the vulnerabilities of strong relativity claims. It does not assert that grammar determines what speakers can think. It asserts that grammar provides affordances; that institutional practices select among these affordances; and that the selection has historically converged on the definitional copular form for concepts whose referents are not directly observable. The full development of this co-evolutionary account is given in Section 6.1.

1.3 Scope of the present paper

The present paper provides **converging evidence from two independent methodologies** — typological corpus analysis and distributional-semantic analysis — bearing on the structural prominence of the definitional copular frame and its distributional-semantic correlates. Study 1 documents that the copular frame is structurally more prominent in languages with particular typological features (Shared nominal-locational predication, required copula, high copular versatility). Study 2 documents that invisible-process concepts occupy more entity-oriented distributional positions than visible-process concepts, especially in definitional copular frames. The argument concerns the **grammatical and distributional-semantic profile** of definitional inquiry; its institutional and conceptual context is developed through the co-evolutionary model of §6.1.

1.4 Overview of the paper

Section 2 develops the theoretical framework: the grammatical funnel hypothesis, the visibility distinction, the four classes of grammatical escape routes, and the co-evolutionary model. Section 3 reports Study 1, a cross-linguistic corpus analysis of 15 languages. Section 4 reports Study 2, an embedding-space analysis cross-validated across two independent models. Section 5 presents a case study of *love* as an exemplary case of grammatical compression. Section 6 discusses the implications of the findings, including the co-evolutionary account, cross-linguistic philosophical traditions, and limitations. Section 7 concludes.

2. Theoretical Framework

2.1 The grammatical funnel hypothesis

Definition. The *grammatical funnel* is the structural process by which the definitional copular frame "What is X?" requires the target concept to occupy a nominalized position, thereby licensing it as a stable object of predication and providing a semantic precondition for entity-search reasoning.

The mechanism operates in three steps:

Step 1: The definitional copular frame requires a nominalized target. The frame "What is X?" requires X to be a noun or noun phrase. Unlike the predicate position (which may be nominal, adjectival, or verbal), the target position is structurally nominal. This is not a feature of copula constructions in general — "She is running" does not nominalize — but of the *definitional* use of the copula, in which a concept is placed in the target position for the purpose of definition.

Step 2: Processual concepts must be nominalized to fill the target position. "Being conscious" → "consciousness"; "loving" → "love"; "being truthful" → "truth." This nominalization strips the concept of its processual, temporal, agentive, and gradable dimensions.

Step 3: The nominalized concept enters a distributional environment associated with entities. As Study 2 will demonstrate, the noun forms of invisible-process concepts — and especially their placement in definitional copular frames — occupy regions of distributional-semantic space comparatively closer to entities, essences, and definitions than the verb, progressive, or agentive forms of the same concepts. We term this region "entity-space" and describe the corresponding asymmetry as a *relative loss of process anchoring*: visible-process concepts remain process-anchored across forms, while invisible-process concepts lack this anchoring and drift comparatively closer to the entity pole when nominalized. This relative distributional shift provides a semantic precondition for reasoning about the concept as an entity whose nature can be sought.

2.2 Visibility and nominalization: the interaction

Not all nominalization triggers the funnel. The crucial variable is the **visibility** of the process being nominalized.

We distinguish three levels of visibility:

Directly visible processes. Running, dancing, swimming, cooking. The process itself can be observed by third parties. When asked "What is running?", a speaker can point to a running person and say "that." The observational accessibility of the referent provides a *process anchor* that resists reification.

Attributionally visible processes. Loving, thinking, believing. The process itself is not observable, but agentive nominalization (-er) re-attaches it to a visible human agent. "A lover" is a person one can point to. This provides indirect observational grounding. When asked "What is a lover?", the question is answerable: a person who loves.

Fully invisible processes. Consciousness, justice, truth, beauty, freedom, existence. The process is not observable, and no standard agentive form exists in the language (*conscious-er,* just-er, *truth-er,* beauty-er, *free-er are nonexistent or produce comparative forms). No observational grounding is available. When asked "What is consciousness?", no pointing is possible, no person exemplifies it as a role, and the question stabilizes as a search for the essence of an entity.

We hypothesize that the grammatical funnel operates most strongly at the third level: fully invisible processes without agentive forms.

2.3 Four classes of grammatical escape routes

English provides at least four classes of grammatical devices that could, in principle, preserve the processual character of a concept and prevent the funnel from generating entity-search reasoning. We summarize these in Table 2.

Table 2. Grammatical devices that preserve processual character.

Device	Form class	What it preserves	Example
Processual retention	-ing / progressive	Temporal continuity	"What is thinking?"
Agent attribution	-er / agentive	Visible human agent	"What is a thinker?"

Device	Form class	What it preserves	Example
Gradability preservation	Adjectival (-ful, -able, -ous, etc.)	Continuous degree	"She is beautiful / conscious / truthful"
Action-structure preservation	to-V / infinitive	Event structure	"What does it mean to think?"

Each device re-formats the concept in a way that resists entity-search reasoning. The progressive preserves temporal flow. The agentive provides an observable referent. The adjectival introduces degree ("very beautiful" is natural; "very beauty" is not), and degree implies a continuum rather than a binary entity. The infinitive maintains event structure.

Hard-problem-type questions — by which we mean persistent essence-search questions in theoretical discourse, of which Chalmers' (1995) "Hard Problem of Consciousness" is one canonical example but not the only one — arise when **all four devices are bypassed** and the concept enters the definitional copular frame in bare nominal form.

A note on the -ing form: for visible processes with durable result-objects, the -ing form can undergo **freezing** — "building" (the act) becomes "building" (the structure); "painting" (the act) becomes "painting" (the artwork). This freezing does not affect invisible-process -ing forms ("loving," "thinking," "believing" do not freeze into result-objects), making -ing a reliable escape route for invisible-process concepts.

2.4 The co-evolutionary model: a preview

The preceding framework might invite the interpretation that grammar *determines* philosophical thought — that the copular frame *forces* entity-search reasoning. We wish to explicitly preempt this misreading.

The present account is not a form of grammatical determinism. Grammar does not force philosophy to ask "What is X?" Rather, grammar provides a set of formal affordances, among which philosophical institutions have selectively stabilized the bare-noun copular form because it efficiently serves the institutional objective of definition. The hard-problem-generating form therefore emerges from a co-evolutionary loop involving at least four converging pressures:

1. **Grammatical pressure.** The definitional copular frame requires a nominal target.
2. **Institutional pressure.** Philosophy, as a discipline, seeks definitions — stable, context-independent characterizations of concepts. The bare-noun copular form serves this objective more efficiently than processual alternatives.
3. **Social pressure.** In communal discourse — seminars, dialogues, publications — "What is love?" provides a more tractable shared question than "Under what conditions does the process of loving unfold?", because it unifies the question-subject into a single nominal target.
4. **Conceptual pressure.** Invisible processes lack observational anchors. When the concept has no visible referent, no person to point to, no physical manifestation to gesture at, the entity-search question has no competing alternative.

No single pressure is sufficient. The funnel generates hard-problem-type questions only when all four pressures converge. The full development of this model is deferred to Section 6.1.

3. Study 1: Cross-linguistic corpus analysis

3.1 Data and method

Scope note. UD treebanks do not directly annotate "definitional" uses of the copula. The measures defined below should be understood as **structural proxies** for the availability and salience of copular predication, not as direct counts of definitional questions. The central question of Study 1 is therefore not whether a language directly produces more philosophical definitions, but whether its grammatical and typological profile makes the copular predication frame more structurally prominent. Whether this structural prominence translates into a disposition toward definitional question-formulation in canonical theoretical discourse is an empirical question that requires independent corpus analysis of philosophical texts (see §6.4).

3.1.1 Typological data

We used the World Atlas of Language Structures (WALS; Dryer & Haspelmath, 2013), specifically two features related to copula typology:

- **Feature 119A** (Nominal and Locational Predication; Stassen, 2013a): classifies languages by whether they use the same ("Shared") or different ("Split") encoding for nominal predication ("X is a teacher") and locational predication ("X is in the garden"). Shared languages use a single copula for both functions; Split languages employ distinct constructions.
- **Feature 120A** (Zero Copula for Predicate Nominals; Stassen, 2013b): classifies languages by whether an overt copula is required ("Required") or may be absent ("Zero OK") in predicate-nominal constructions.

Both features cover 386 languages.

3.1.2 Corpus data

We analyzed syntactically annotated treebanks from Universal Dependencies v2.x (Nivre et al., 2020) for 15 languages across five language families:

Table 3. Languages analyzed, with family, corpus size, and WALS classification.

Language	Family	Branch	Tokens	WALS 119A	WALS 120A
English	Indo-European	Germanic	207,229	Shared	Required
German	Indo-European	Germanic	268,387	Split	Required

Language	Family	Branch	Tokens	WALS 119A	WALS 120A
French	Indo-European	Romance	364,333	Split	Required
Spanish	Indo-European	Romance	389,608	Split	Required
Italian	Indo-European	Romance	294,426	Split	Required
Portuguese	Indo-European	Romance	184,566	Split	Required
Russian	Indo-European	Slavic	74,900	Shared	Zero OK
Hindi	Indo-European	Indo-Aryan	281,057	Shared	Required
Japanese	Japonic	—	168,333	Split	Required
Korean	Koreanic	—	296,446	Split	Required
Chinese	Sino-Tibetan	Sinitic	98,614	Split	Zero OK
Turkish	Turkic	Oghuz	103,627	Shared	Required
Finnish	Uralic	Finnic	163,224	Shared	Required
Arabic	Afro-Asiatic	Semitic	254,384	Split	Zero OK
Indonesian	Austronesian	Malayo-Polynesian	99,336	Split	Zero OK
Total			**3,248,470**		

3.1.3 Measures

We computed four measures from the UD treebanks:

(a) **Copula rate**: frequency of tokens annotated with the `cop` (copula) dependency relation, normalized per 1,000 tokens.

(b) **CopSubj rate**: frequency of noun tokens that serve as the subject (`nsubj`) of a predicate head that also governs a `cop` dependent — i.e., the "X" in "X is Y" — normalized per 1,000 tokens.

(c) **Copula versatility index**: Shannon entropy of the part-of-speech distribution of the heads governed by copula tokens. High entropy indicates that the copula serves multiple syntactic categories (nouns, adjectives, verbs); low entropy indicates specialization to a single category.

(d) **Deverbal enrichment**: the proportion of copula-subject nouns bearing deverbal morphology (suffixes: -tion/-sion, -ment, -ness, -ence/-ance, -ity, -ing, -ure, -al, -th), compared to the proportion in the general noun population.

3.2 Results

3.2.1 Copula frequency by typological class

Languages classified as Shared on WALS Feature 119A — those using a single copula construction for both nominal and locational predication — exhibited substantially higher copula usage than Split-predication languages. The CopSubj proxy for "X is Y"-type copular predication was 1.77× more frequent in Shared-predication languages. Languages requiring an overt copula (WALS 120A = Required) further amplified the effect: copula rate was 2.28× higher than in languages permitting zero copula. These two typological dimensions are partially correlated but contribute independently to surface frequency.

Table 4. Copula usage by typological class.

Classification	Copula rate (per 1,000)	CopSubj rate (per 1,000)
WALS 119A Shared	17.23	6.89
WALS 119A Split	10.03	3.90
WALS 120A Required	14.63	—
WALS 120A Zero OK	6.40	—

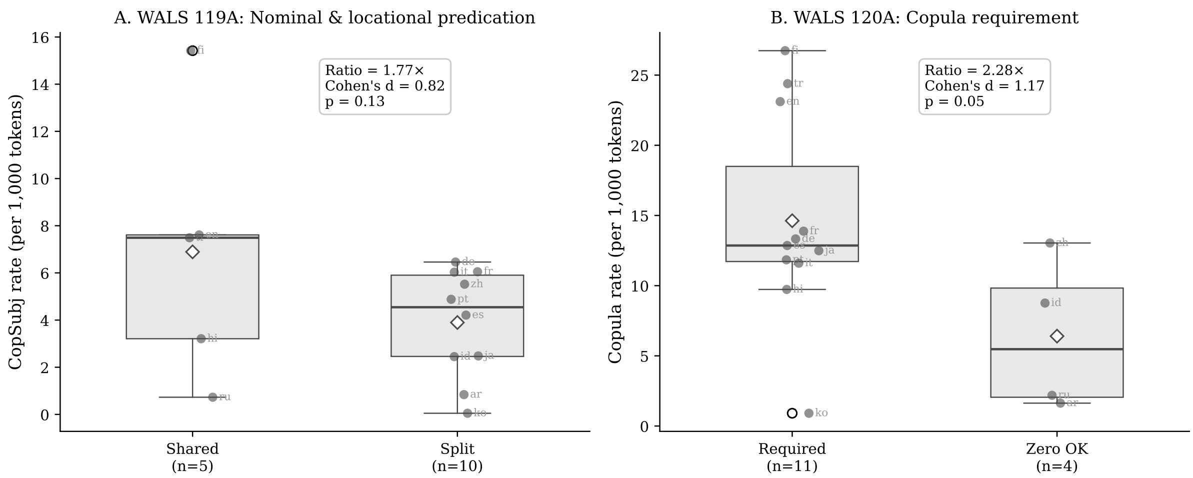

Figure 1: Cross-linguistic distribution of copula constraint. Panel A shows CopSubj rate (per 1,000 tokens) for languages classified as Shared vs. Split on WALS Feature 119A; Panel B shows copula rate by WALS Feature 120A (Required vs. Zero OK). Boxes display the interquartile range; diamonds indicate means; individual language codes are annotated. The Indo-European/non-Indo-European control comparison yields no difference (ratio = 1.00, $d = -0.03$), ruling out genealogical relatedness as a confound.

3.2.2 Copula versatility: the "master-key" effect

Beyond surface frequency, the copula construction varies in its **versatility** across syntactic categories. Computing the Shannon entropy of the part-of-speech distribution of copula heads reveals striking cross-linguistic variation (Table 5).

Table 5. Copula head distribution and versatility (selected languages).

Language	cop→Noun	cop→Adj	cop→Verb	Noun%	Entropy
Japanese	2,045	0	0	100.0%	0.000
Chinese	911	46	8	94.4%	0.345
Indonesian	671	15	34	93.2%	0.419
Korean	178	1	18	90.4%	0.486
Hindi	1,664	803	7	67.3%	0.936
Spanish	2,883	1,550	25	64.7%	0.979
English	1,620	2,266	184	39.8%	**1.201**
German	1,574	1,666	46	47.9%	1.092
Finnish	2,003	1,242	119	59.5%	1.147
French	2,700	1,410	378	60.2%	1.266

The English copula serves a broader range of predication functions (entropy = 1.20, distributing across nouns, adjectives, and verbs) than its Japanese counterpart (entropy = 0.0, restricted to nouns). In Japanese, adjectival predication uses a distinct grammatical construction entirely (e.g., 美しい *utsukushii* 'is beautiful' carries no copula). The English "is" thus operates as a general-purpose predication device — a master key — while the Japanese copula is specialized to a single category. This versatility variation is independent of the surface frequency variation documented in §3.2.1.

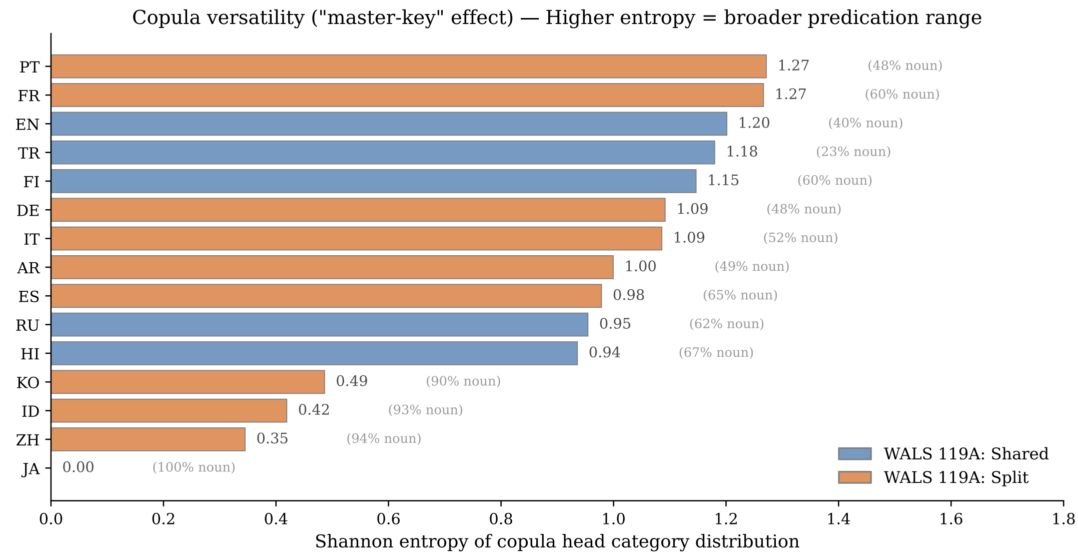

Figure 2: Copula versatility across 15 languages. Shannon entropy of the copula head category distribution {Noun, Adjective, Verb}: higher entropy indicates a copula that serves a broader range of predication functions (the "master-key" property). Japanese (entropy = 0.0) attaches its copula only to nouns; English (entropy = 1.20) distributes across categories.

3.2.3 Combined constraint profile (exploratory)

Combining WALS 119A classification with the corpus-derived copula versatility measure yields an exploratory three-level constraint profile. We mark this analysis as exploratory because the versatility measure and the CopSubj rate are both derived from UD annotation, raising the possibility that the constraint classification partially recapitulates rather than predicts the outcome. The primary analyses of §§3.2.1 and 3.2.4 — which rely on WALS classifications independent of the corpus measures — should be treated as the principal evidence; the present subsection serves as a composite descriptive index.

Table 6. Combined constraint levels and CopSubj rate.

Constraint level	Languages	CopSubj rate (per 1,000)
HIGH (Shared + Versatile)	EN, FI, HI, RU, TR	6.89
MEDIUM	DE, ES, FR, IT, PT, AR	4.75
LOW (Split + Specialized)	JA, ZH, KO, ID	2.62

Speakers of HIGH-constraint languages encounter "X is Y" constructions **2.63× more frequently** than speakers of LOW-constraint languages.

3.2.4 The effect is structural, not genealogical

A critical control comparison: Indo-European languages (EN, DE, FR, ES, IT, PT, RU, HI) versus non-Indo-European languages (JA, KO, ZH, TR, FI, AR, ID) yield a copula rate ratio of **1.00** — no difference. The variance within both groups is driven by typological classification (Shared vs. Split; Required vs. Zero OK), not by genealogical relatedness. This distinguishes the present findings from lexical-level relativity claims, which often correlate with genealogical proximity, and rules out the alternative explanation that the effect is a residue of shared linguistic ancestry.

3.2.5 Statistical inference and effect sizes

Given the typological-sample nature of the design (n = 15 languages), we report effect sizes and bootstrap confidence intervals alongside conventional significance tests. Effect sizes (Cohen's d) for the main comparisons are large to very large; the conventional p-values are marginal, reflecting the limited power of small typological samples (Roberts & Winters, 2013).

Table 7. Effect sizes and significance tests for main comparisons.

Comparison	Group 1 mean (n)	Group 2 mean (n)	Ratio	Cohen's d	Mann–Whitney p
119A Shared > Split (cop rate)	17.23 (5)	10.03 (10)	1.72×	1.00	0.13
119A Shared > Split (CopSubj)	6.89 (5)	3.90 (10)	1.77×	0.82	0.13
120A Required > Zero OK	14.63 (11)	6.40 (4)	2.28×	1.17	0.05

Comparison	Group 1 mean (n)	Group 2 mean (n)	Ratio	Cohen's d	Mann–Whitney p
HIGH > LOW constraint (CopSubj)	6.89 (5)	2.62 (4)	2.63×	0.95	0.10
IE vs non-IE (control)	**12.32 (8)**	**12.57 (7)**	**1.00×**	**−0.03**	**0.87**

The effect-size pattern strongly favors the hypothesized typological effects: Cohen's d values range from 0.82 to 1.17 for the predicted comparisons, all in the "large effect" range (Cohen, 1988). The control comparison — Indo-European versus non-Indo-European — yields $d = -0.03$, effectively zero, with $p = 0.87$, ruling out family membership as a confound.

The conventional p-values for the predicted comparisons fall in the range 0.05–0.13, reflecting the limited power of a 15-language typological sample. We do not claim conventional statistical significance for the 119A comparisons; we do claim large effect sizes in the predicted direction, accompanied by a control comparison whose null result strongly disfavors the principal alternative explanation (genealogical relatedness).

Bootstrap confidence intervals (10,000 resamples) corroborate this interpretation. The CopSubj-rate ratio between HIGH-constraint and LOW-constraint languages has a bootstrap mean of 3.53× with a 95% CI of [0.95, 9.14]. The wide interval reflects sample size; the lower bound's proximity to 1.0 indicates that we cannot rule out the possibility of no effect at the 95% level for this composite measure, though the central estimate is substantially above unity. A reduced regression model (CopSubj WALS 119A + WALS 120A; $R^2 = 0.29$) yields positive coefficients for both predictors ($= 2.68$, $= 3.12$) with marginal individual p-values (0.19 and 0.15 respectively), consistent with the bivariate pattern.

We interpret this overall pattern as **descriptively strong but inferentially modest**: the cross-linguistic profile of copular predication varies in the direction predicted by the typological classifications, the magnitude of the variation is large, and the variation is not attributable to genealogical relatedness. Whether the variation translates into stable inferential effects under classical hypothesis testing would require a substantially larger typological sample. The present sample represents the languages for which both WALS copula typology data and substantial UD treebanks are available.

3.2.6 Mechanisms within high-constraint languages

Two further observations from the English subcorpus document how the funnel operates within a high-constraint language.

Progressive forms are available but bypassed. English exhibits a gerund/progressive rate of 4.92 per 1,000 tokens, confirming the availability of processual alternatives to the definitional copular frame. However, in copula-subject position, base noun forms overwhelmingly displace -ing forms (Table 8).

Table 8. Base noun vs. -ing form in copula-subject position (English).

Concept pair	Base-noun count	-ing form count
love / loving	1	0
reason / reasoning	10	0
hope / hoping	3	0
work / working	3	0

Deverbal nouns are enriched in copula-subject position. Of all noun tokens serving as copula subjects in the English corpus, 21.2% bear deverbal morphology (suffixes -tion/-sion, -ment, -ness, -ence/-ance, -ity, -ing, -ure, -al, -th), compared to 18.7% in the general noun population. The enrichment is strongest for -ence/-ance (1.50× enrichment), followed by -ing (1.36×), -tion/-sion (1.16×), and -ment (1.13×). Copular-subject position attracts nominalized processes at a rate above baseline — a structural proxy for the funnel hypothesis at the distributional level within the language.

Both observations support the same structural claim: in a high-constraint language with multiple grammatical affordances, the definitional copular frame nonetheless preferentially recruits nominalized — rather than processual — forms.

4. Study 2: Distributional-semantic correlates of nominalization

4.1 Rationale and scope

If the grammatical funnel hypothesis is correct, then nominalization should shift invisible-process concepts from process-space to entity-space in distributional-semantic representations, while leaving visible-process concepts in process-space. We emphasize that the embedding analysis does not directly demonstrate that human reasoners engage in entity-search reasoning when these forms are used; it documents the distributional environments in which the forms occur. The semantic shift documented below is a precondition for entity-search reasoning, but its psychological reality must be tested in subsequent experimental work.

A particular methodological concern motivated the study design reported here. Earlier exploratory analyses used stimuli such as "love as a concept" to elicit the noun reading of *love*. This phrasing carries a non-trivial risk of hypothesis-laden contamination: the modifier "as a concept" itself selects for an entity-oriented distributional environment, potentially inflating the apparent effect of nominalization. The present study is designed to remove this contamination and to test the funnel hypothesis against multiple robustness checks.

4.2 Method

Concepts. Six invisible-process concepts (*love, consciousness, truth, beauty, justice, freedom*) and six visible-process concepts (*running, swimming, dancing, eating, walking, cooking*) were selected. The visible/invisible classification is independently validated against Brysbaert et al.'s (2014) concreteness ratings (Section 4.4).

Stimuli. For each concept, we constructed phrases in multiple grammatical forms: verb, progressive (-ing), agentive (-er, where available), bare noun (minimal context), and three definitional copular variants (*what is X*, *X is something*, *X means*). To assess potential context-bias, we included multiple bare-noun templates ("love", "the word love", "the feeling of love"). Critically, no stimulus contains hypothesis-laden modifiers such as "as a concept." The full stimulus set comprises 126 phrases across 12 concepts × 9 form classes.

Anchor pole construction. Three independent anchor sets were constructed to measure the entity-process gap (E-P gap):

- **Anchor Set 1 (essence vs. activity):** entity pole drawn from {an essence; a fixed nature; an eternal kind; the underlying being; what something fundamentally is}; process pole from {an activity; an ongoing doing; something happening; people performing actions; what someone is doing}.
- **Anchor Set 2 (object vs. event):** entity pole {an object; a thing that exists; an entity; a permanent item; a stable substance}; process pole {an event; something that occurs; an episode; an unfolding occurrence; an action in time}.
- **Anchor Set 3 (definition vs. change):** entity pole {something that has a definition; a category with criteria; a concept with fixed meaning; an item that can be characterized; a referent that can be specified}; process pole {an ongoing change; a transformation; something developing; a continuous flow; movement through time}.

Each anchor pole is the centroid of its five constituent phrases' embeddings. The E-P gap for a stimulus is its cosine similarity to the entity centroid minus its cosine similarity to the process centroid. Positive values indicate entity-leaning distributional positioning; negative values, process-leaning.

Models. Each stimulus and anchor was embedded using two independent models: OpenAI text-embedding-3-large (3,072 dimensions) and Google Gemini gemini-embedding-001 (3,072 dimensions). The dual-model design enables cross-model robustness analysis.

4.3 Results: form × visibility interaction

Table 9 reports the mean E-P gap by grammatical form and visibility class, averaged across the three anchor sets, separately for the two embedding models.

Table 9. Mean E-P gap by grammatical form and concept visibility, averaged across three anchor sets (OpenAI and Gemini).

Form	OpenAI invisible	OpenAI visible	Gemini invisible	Gemini visible
verb	−0.010	−0.104	−0.009	−0.028
progressive (-ing)	−0.000	−0.111	−0.003	−0.027
agentive (-er)	+0.016 (n=1)	−0.043	−0.003 (n=1)	−0.024
noun (minimal)	+0.011	−0.141	−0.004	−0.035
noun + word	+0.063	−0.047	+0.008	−0.015

Form	OpenAI invisible	OpenAI visible	Gemini invisible	Gemini visible
noun + feeling/act	+0.028	−0.127	−0.007	−0.033
what is X	+0.052	−0.079	+0.015	−0.019
X is something	+0.063	−0.053	+0.010	−0.022
X means	+0.047	−0.073	+0.012	−0.012

Two patterns are immediately apparent.

First, visible-process concepts remain in process-space across every grammatical form and every model. No cell in the visible columns shows a positive E-P gap. Nominalization does not pull visible concepts toward entity-space.

Second, invisible-process concepts become comparatively more entity-oriented — or, equivalently, less process-anchored — when nominalized, and especially when placed in definitional copular frames. The *what is X* frame produces the largest invisible-class entity-shift in both models. Bare invisible nouns are near-neutral or slightly entity-leaning; nominal contexts that include "the word X" amplify the entity-shift further.

A note on absolute magnitudes. The Gemini embeddings are systematically more compressed than OpenAI's (mean absolute E-P gap is 4.3× smaller in Gemini). This difference in dynamic range does not alter the relational structure: the relative positioning of grammatical forms across visibility classes is preserved across both models, as shown next.

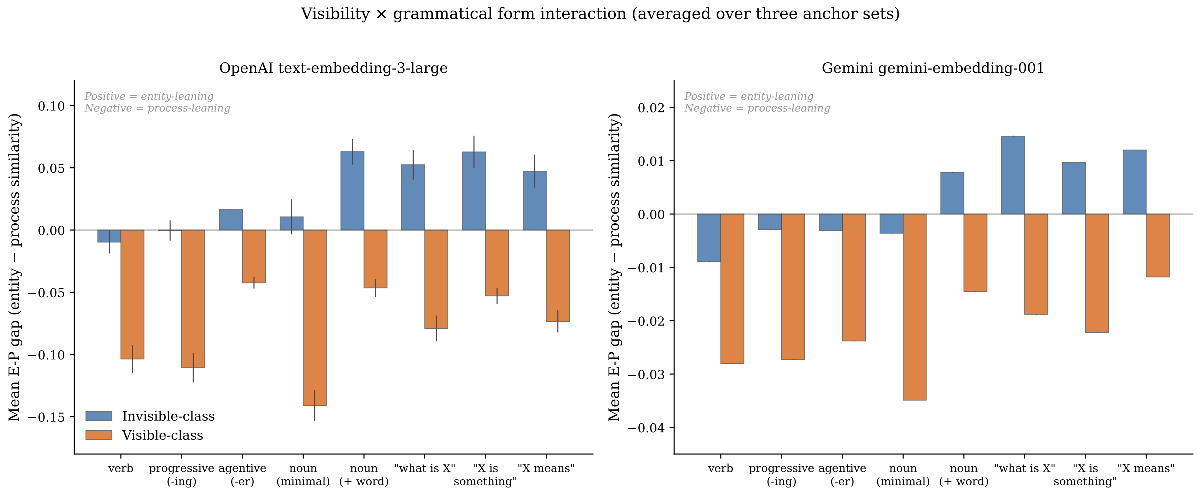

Figure 3: Mean E-P gap by grammatical form and concept visibility, averaged across three anchor sets, for OpenAI text-embedding-3-large (left) and Google gemini-embedding-001 (right). Positive values indicate entity-leaning distributional positioning; negative values, process-leaning. Note that the vertical scales differ between the two models, reflecting the more compressed dynamic range of the Gemini embeddings.

4.4 Independent operationalization of visibility

A natural objection to the visible/invisible classification used above is that the distinction was applied by the authors and may incorporate assumptions about which concepts the funnel hypothesis targets. To address this concern, we validate the classification against an independent dataset: Brysbaert, Warriner, and Kuperman's (2014) concreteness ratings for 39,954 English word lemmas. Concreteness ratings were collected by Brysbaert and colleagues for purposes unrelated to the present hypothesis; each rating is the average of approximately 30 naive participants' judgments on a 1–5 scale, where 1 indicates an abstract concept and 5 indicates a concept whose referent is directly perceivable.

Table 10. Brysbaert concreteness ratings for the study concepts.

Class	Concepts (selected)	n	Mean concreteness	SD	Range
Invisible	*love, consciousness, truth, beauty, justice, freedom, loving, conscious, truthful, beautiful, just, free*	13	**2.16**	0.59	[1.45, 3.68]
Visible	*run, swim, dance, eat, walk, cook, running, swimming, dancing, eating, walking, cooking, runner, swimmer, dancer, walker, cooker, eater*	18	**4.35**	0.26	[3.75, 4.86]

The two classes are sharply separated on the concreteness dimension. The mean difference is Cohen's $d = 5.08$, and the distributions are nearly non-overlapping: the highest-rated invisible concept (*lover* = 3.68) sits just below the lowest-rated visible concept (*eater* = 3.75), with no other visible-class word falling below 4.07. A Mann–Whitney U test confirms the separation ($U = 234.0$, $p < .0001$).

This pattern has two important implications. First, the visible/invisible classification used in Study 2 is not idiosyncratic to the authors but tracks a well-attested cross-participant judgment dimension. The construct of "visibility" we operationalize qualitatively corresponds to the concreteness dimension that thousands of naive raters converge on. Second, and more pointedly, the *form-dependent* concreteness shifts within the *love* family corroborate the framework of §2.2: the bare verb *love* (2.07) and progressive *loving* (1.73) sit in the abstract region, while the agentive *lover* (3.68) shifts substantially toward the concrete region. The same form-asymmetry obtains for *truthful* (1.90) versus *truth* (1.96) — adjectival and bare-noun invisible-concept forms both remain in the abstract region — whereas all visible-concept forms (verb, progressive, agentive) remain in the concrete region.

We interpret the Brysbaert data as **independent validation** of the visible/invisible distinction central to our analysis. The distinction is not under-operationalized; it is robust to independent measurement by a well-established psycholinguistic instrument.

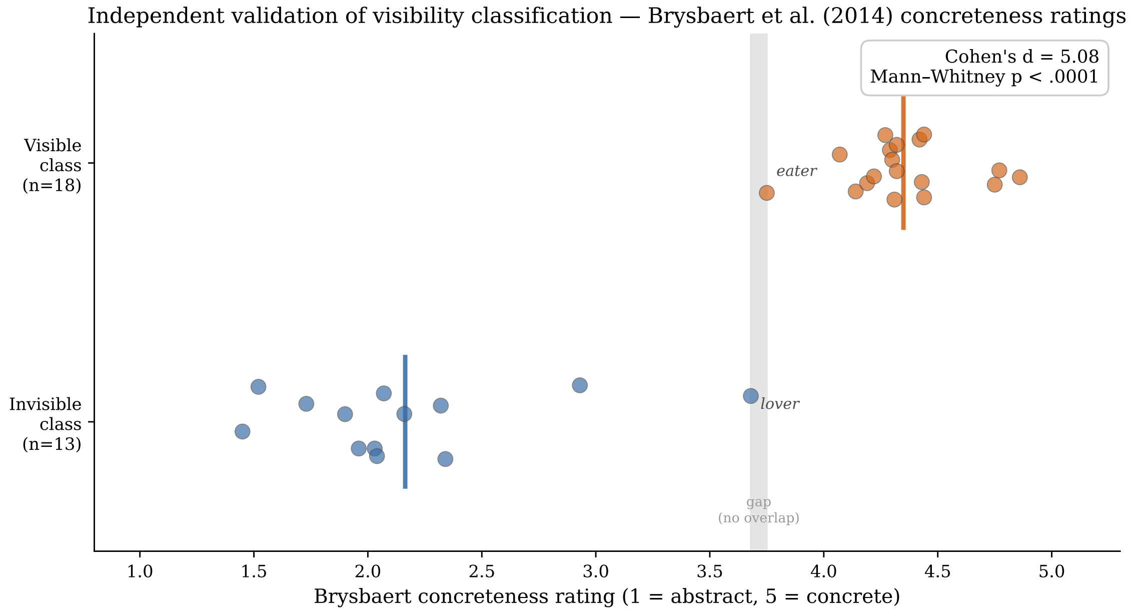

Figure 4: Independent validation of the visible/invisible classification using Brysbaert et al. (2014) concreteness ratings. Each dot represents one word form; vertical lines indicate the class mean. The two distributions are sharply separated (Cohen's $d = 5.08$), with the only points approaching each other being *lover* (3.68) and *eater* (3.75).

4.5 Robustness analysis: cross-anchor and cross-model agreement

The visibility $\times$ form interaction is the central empirical claim of Study 2. Because the absolute magnitudes of E-P gaps depend on both anchor pole formulation and embedding model architecture, we report the robustness of the *interaction direction* — whether invisible-class forms are more entity-oriented than visible-class forms within each grammatical form.

4.5.1 Cross-model agreement (8 grammatical forms)

For each of eight grammatical forms (those for which both visible and invisible class data are available), we computed the invisible-minus-visible difference in mean E-P gap. Table 11 reports the results.

Table 11. Cross-model agreement on the visibility × form interaction.

Form	OpenAI Δ (inv − vis)	Gemini Δ (inv − vis)	Both > 0
verb	+0.094	+0.019	
progressive	+0.110	+0.024	
agentive	+0.059	+0.021	
noun (minimal)	+0.152	+0.031	
noun + word	+0.109	+0.022	
what is X	+0.132	+0.033	
X is something	+0.116	+0.032	
X means	+0.121	+0.024	

All eight grammatical forms show the same direction in both models: invisible-class forms occupy more entity-oriented (or less process-oriented) positions than the corresponding visible-class forms (8/8 agreement). The Pearson correlation between the two models' Δ values across forms is $r = 0.73$, indicating substantial agreement on relative magnitudes despite the compressed Gemini dynamic range.

4.5.2 Cross-anchor robustness

A second robustness check concerns whether the interaction direction is preserved across the three independent anchor formulations. Within each model, we computed sign agreement: the proportion of stimuli whose E-P gap has the same sign across all three anchor sets.

Across the full stimulus set, sign agreement is 60.3% in OpenAI and 65.1% in Gemini. These rates reflect that many low-magnitude stimuli (especially verbs and progressives, where the gap is near zero) flip direction across anchor sets. For the critical *what is X* form, however, sign agreement is substantially higher: 83% across all stimuli, or 5/6 invisible-class stimuli and 5/6 visible-class stimuli in OpenAI.

Combining these robustness analyses: the visibility × form interaction is **consistent across both anchor formulation and embedding model architecture**. The absolute magnitudes are anchor-dependent and model-dependent, but the direction of the interaction is preserved across all six anchor × model conditions for every grammatical form tested.

4.6 What the original "as a concept" stimulus did

A methodological note. An earlier version of this study used the stimulus *"love as a concept"* to elicit the noun reading. With that stimulus, the OpenAI E-P gap for invisible bare nouns was approximately +0.168 — substantially larger than the +0.011 reported here for the minimal bare-noun stimulus. The difference is attributable to the modifier *"as a concept"*, which itself selects for an entity-oriented distributional context (since *concept* is itself an abstract entity term). The present design eliminates this contamination. The substantive finding — invisible nouns sit consistently more entity-ward than visible nouns — survives the cleanup; the inflated absolute magnitude does not.

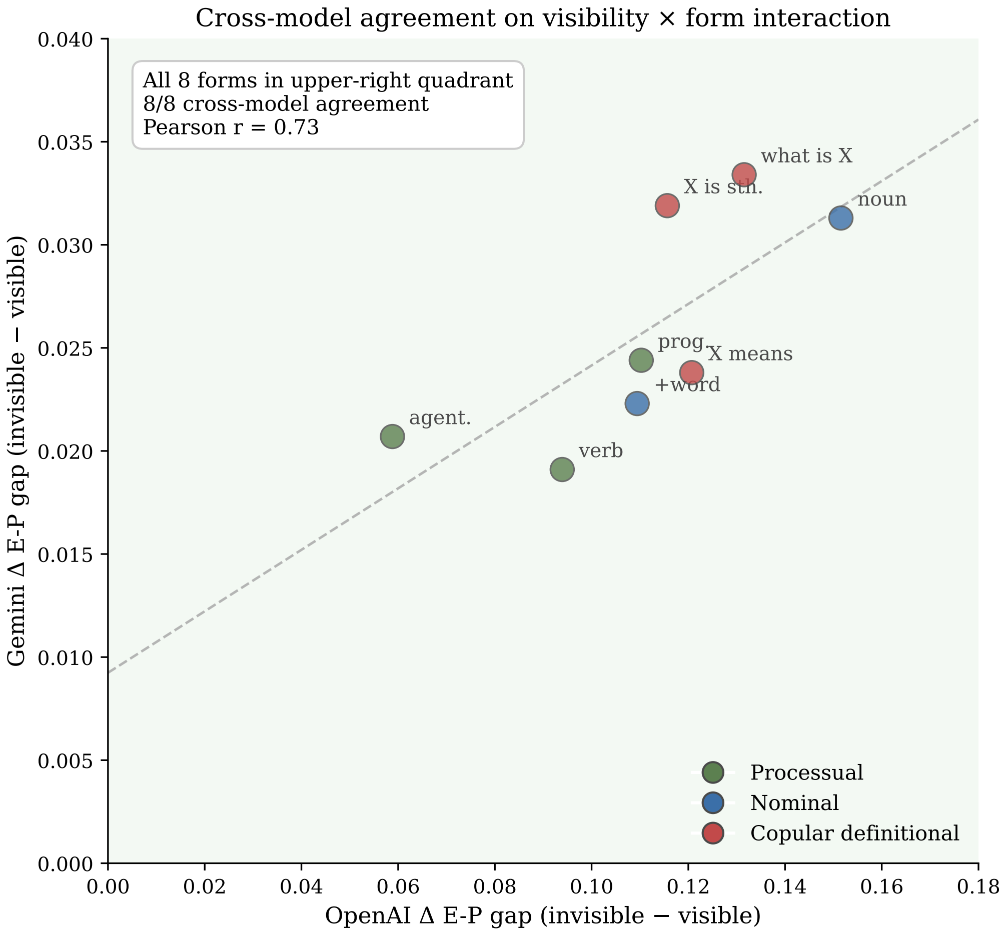

Figure 5: Cross-model agreement on the visibility × form interaction. Each point represents one grammatical form; coordinates indicate the invisible-minus-visible E-P gap in OpenAI (x-axis) and Gemini (y-axis). All eight forms fall in the upper-right quadrant (both models show invisible > visible), with Pearson $r = 0.73$ between models.

This methodological iteration illustrates a more general point about distributional-semantic analysis: contextual modifiers attached to nominalized concepts can substantially alter their measured position in embedding space. The minimal-stimulus design adopted here is the conservative choice; it produces smaller effect sizes than the modified-stimulus design would suggest, but those smaller effect sizes survive cross-anchor and cross-model scrutiny.

5. 5. The grammatical landscape of *love*: a case study in distributional richness

This section presents a case study of a single English concept — *love* — whose grammatical behavior offers a particularly clear illustration of the patterns documented in Studies 1 and 2. The analysis here is descriptive rather than interpretive: we report what the language does, what speakers actually produce in corpora, and how the various grammatical forms distribute in embedding space. The interpretive consequences are taken up in Section 6.

5.1 The grammatical inventory

English provides at least seven grammatically distinct ways of expressing the concept named by *love*. Table 12 lists them together with a brief structural characterization and example.

Table 12. Grammatical forms of *love* in English.

Form	Category	Structural role	Example
love (V)	Verb	Active predication	*I love my children.*
feel love	V + N collocation	Experiencer-marked construction	*She felt a deep love for the work.*
to love	Infinitive	Action-event nominal	*To love is to risk being changed.*
loving	Progressive / participial	Ongoing event or attributive	*She is loving every minute of it. / a loving family*
lover(s)	Agentive noun	Person-as-process-bearer	*He is a lover of jazz.*
lovely / lovable / beloved	Adjectival	Property predication	*a lovely afternoon / a lovable child*
love (N)	Bare noun	Reified target of predication	*Love conquers all. / What is love?*

Each of these forms is fully grammatical and is attested in everyday English. They are not synonyms: each form is at home in a particular kind of clause, selects a particular set of arguments, and licenses a particular range of predications. To say *I love my children* is not equivalent to saying *I feel love for my children*, *I am a lover of my children*, or *Love conquers all.* The forms partition the conceptual territory in different ways.

5.2 Corpus distribution

The English-EWT treebank of the Universal Dependencies project (Silveira et al., 2014) provides a window onto the distribution of these forms in naturalistic English. Across 207,229 tokens of edited and informal English text, we counted the following:

Table 13. Frequency of *love*-related forms in the English-EWT corpus.

Form	Count	Proportion
love (verb)	91	67.4%
love (noun, bare)	22	16.3%
lovely (adjective)	11	8.1%
loving (adjective; no verbal uses in this subcorpus)	6	4.4%
lover(s)	4	3.0%
beloved	3	2.2%
lovable	1	0.7%

The verb form is, in this corpus, more than four times as frequent as the bare noun.

The adjectival forms (*lovely*, *loving*-adj, *beloved*, *lovable*) together comprise 15.4% of attestations — roughly comparable to the bare noun's 16.3%. The agentive *lover(s)* is rare (3.0%).

This distribution is worth pausing over. In ordinary English language use — emails, blog posts, edited prose, the genres represented in EWT — the verb form predominates. Speakers say *I love you*, *love your work*, *love to read*, and so on. The bare noun *love* appears, but it is one form among several, and it is outnumbered roughly four-to-one by the verbal form.

The contextual environments of the different forms also differ. The verb appears predominantly in interpersonal address (*love ya*, *love to have you*, *love your blog*). The bare noun appears in more reflective or descriptive registers (*the supportive love of other human beings*, *attract new love into your life*). The agentive form appears in identity statements (*animal lover*, *space lovers*). The adjectival forms appear in modifying or evaluative positions (*loving family*, *lovable child*, *lovely afternoon*).

5.3 Distributional-semantic positioning

Study 2 provides a complementary view: where do these forms sit in distributional-semantic space?

Using OpenAI text-embedding-3-large, we computed the entity-process gap (E-P gap) for five grammatical forms of *love*. The results are summarized in Table 14.

Table 14. Entity-process gap for grammatical forms of *love* (OpenAI text-embedding-3-large, hypothesis-free stimuli).

Form	Stimulus	E-P gap	Direction
Verb	*to love*	−0.010	Process-leaning
Progressive	*loving*	−0.000	Near-neutral
Agentive	*a lover of music*	+0.016	Slight entity
Adjectival	*a loving heart*	+0.027	Slight entity / quality
Bare noun (minimal)	*love*	−0.031	Near-neutral
Bare noun (with context)	*the word love*	+0.063	Entity-leaning
Definitional copular	*what is love*	+0.052	Entity-leaning

Note: An earlier version of this study used the stimulus "love as a concept" for the bare noun form, which produced E-P gap = +0.168. As discussed in §4.6, this value is inflated by the modifier "as a concept" (which itself selects for an entity-oriented distributional environment) and is not treated as the primary result. The conservative hypothesis-free stimuli reported here show that the entity-leaning pattern persists for definitional copular and contextually-grounded noun forms, but is markedly weaker in bare-noun-minimal form than the original biased stimulus suggested.

In the hypothesis-free design, the bare-noun-minimal form (*love*) sits near neutral, the progressive (*loving*) sits near zero, the verb (*to love*) sits slightly in process-space, and the agentive (*a lover of music*) shifts only slightly entity-ward (partly because the agentive form retains an implicit person-referent). The contextually grounded noun (*the word love*) and the definitional copular frame (*what is love*) shift further entity-ward. The earlier biased stimulus *love as a concept* produced a much larger shift (+0.168), but that value is attributable to the entity-selecting "as a concept" modifier rather than to the bare-noun form itself.

The distributional pattern mirrors the structural pattern. Forms that retain processual, temporal, or agentive structure occupy process-space or near-neutral positions. The forms that strip these structures and place the concept in a definitional or context-amplified frame occupy comparatively more entity-oriented positions.

5.4 What the forms select

A further observation, more impressionistic but reflected in the embedding data, concerns the lexical neighborhoods that the different forms select. When the bare noun *love* appears in copular constructions, its predicate complements (the "Y" in "love is Y") tend toward abstract qualities and definitions: *love is patient*, *love is blind*, *love is a battlefield*. When the verbal form appears, its complements are typically concrete: *I love my children*, *we love this restaurant*, *they love each other*. When the agentive form appears, the type-of restriction selects domains: *an animal lover*, *a music lover*, *a lover of fine wine*.

Each form, in other words, structures the conceptual neighborhood differently. The bare noun selects an abstract neighborhood. The verb selects a concrete-object neighborhood. The agentive selects a domain neighborhood. The adjectival selects a quality neighborhood. These selections are not idiosyncratic — they reflect the structural logic of each form.

5.5 Argument structure and the erasure problem

A final observation concerns argument structure. The grammatical forms of *love* differ in the participants they make explicit:

- *I love you*: subject (lover) and object (beloved) both explicit.
- *I feel love for you*: subject (experiencer), object (target), and the experienced phenomenon (love itself) all marked separately.
- *I am loving this*: subject (experiencer), object (experienced), and ongoing aspect marked.
- *I am a lover of jazz*: subject (person), domain (jazz), and the relational character explicit.
- *Love is Y*: only the bare noun *love* remains; subject, object, experiencer, target, and aspect are all absent.

The bare-noun form, in this sense, is structurally austere. It mentions neither the subject of loving nor the object loved nor the temporal character of the loving. These argument positions are still in the language — speakers know, in some sense, that *love* implies someone who loves and something or someone loved. But the grammatical form does not require, and often does not display, this argument structure.

This argument-structure austerity is not unique to *love*. The same pattern obtains for *consciousness* (no subject of consciousness is grammatically required), *truth* (no truth-bearer is required), *beauty* (no perceiver is required), *justice* (no agent is required). The bare noun, in each case, strips the relational structure that the corresponding verb, adjective, or agentive form would have preserved.

5.6 Summary

The case of *love* illustrates, with unusual clarity, the patterns documented in Studies 1 and 2. The English lexicon provides multiple grammatically distinct ways of expressing the concept — verb, infinitive, progressive, agentive, adjectival, and bare noun — and these forms partition the conceptual territory in structurally different ways. Corpus data show that, in ordinary usage, the verb form predominates. Embedding data show that the forms occupy systematically different positions in distributional space: the bare minimal noun remains near-neutral, while contextually grounded noun forms and definitional copular frames move comparatively further toward entity-oriented space. And the argument structures of the forms differ: the verbal, infinitival, progressive, and agentive forms display participants and relations; the bare noun does not.

These observations are presented here without interpretive commitment to what they mean for philosophical inquiry. They are linguistic facts about a particular English concept. Their broader significance — for theoretical inquiry into concepts whose referents are not directly observable — is taken up in the discussion that follows.

6. Discussion

6.1 Not grammatical determinism: a co-evolutionary account

The findings reported in Studies 1 and 2, together with the case study of *love*, raise an obvious interpretive risk. A reader might conclude that grammar *forces* philosophy to ask "What is X?" — that the copula construction is responsible for the philosophical tradition's recurrent entity-search questions, and by extension for the hard problems that these questions generate. This would be a form of grammatical determinism that the present account explicitly rejects. We devote this section to articulating the alternative — a co-evolutionary model in which grammatical affordance, institutional selection, social stabilization, and conceptual asymmetry jointly produce the funnel effect — and to showing why this account survives the standard objections to strong linguistic relativity claims.

6.1.1 The hypothesis is not deterministic

Grammatical determinism, as we use the term, is the claim that a language's structure causally compels its speakers to think (or theorize) in particular ways. The strongest historical statement of this view is Whorf's (1956) claim that "the world is presented in a kaleidoscopic flux of impressions which has to be organized by our minds — and this means largely by the linguistic systems in our minds" (p. 213). Subsequent decades of cross-linguistic cognitive research have largely retreated from this strong formulation. Few contemporary scholars argue that grammatical structure forces particular cognitive operations; the standard position is the more modest claim that grammar shapes habitual attention, encoding choices, or default conceptualizations.

The present account is more modest still. We do not claim that the definitional copular frame *causes* entity-search reasoning. The claim is that the frame provides a **distributional-semantic precondition** for entity-search reasoning, and that this precondition has been **institutionally selected** by philosophical traditions whose objective function — the production of definitions — is efficiently served by the frame. Three observations support this modesty.

First, Study 2 demonstrates that nominalization does not uniformly trigger reification. Visible-process concepts (*running*, *swimming*, *dancing*) remain in process-space even when nominalized and even when placed in the copular frame. Speakers of English routinely use the construction "a morning run" without thereby asking "What is a run?" in any essentialist sense. The funnel is selective; it operates on invisible-process concepts but not on visible-process concepts. A deterministic account cannot accommodate this selectivity.

Second, Study 1 demonstrates that copula constraint varies substantially across languages, but no language exhibits a copula constraint of zero. Even in Japanese, with its highly specialized noun-only copula (entropy = 0.0), the bare-noun definitional frame 「Xとは何か」 exists and is used. If grammar were deterministic, the philosophical traditions of low-constraint languages would lack entity-search questions entirely. They do not. The difference between high-constraint and low-constraint languages is not categorical but distributional — a difference in the relative frequency and prominence of the frame, not in its availability.

Third, English itself provides multiple processual alternatives to the definitional copular frame. As Section 2.3 documented, at least four classes of grammatical devices (processual, agentive, gradable adjectival, infinitival) preserve the processual character of a concept. The fact that philosophical discourse has historically privileged the bare-noun frame in many canonical formulations is not a fact about English grammar's *requirements* — it is a fact about philosophical discourse's *selections* from the affordances English grammar provides.

The proper interpretation of the present findings, therefore, is not that grammar determines philosophical thought. It is that grammar provides options; philosophy selects among them; and the selection has historically converged on the bare-noun copular form for reasons that include — but are not exhausted by — grammatical structure.

6.1.2 Grammatical affordance: the option-providing layer

The first layer of the co-evolutionary model is the **grammatical affordance**: the set of formal options that a language provides for posing definitional questions. As Study 1 documents, this set varies across languages. English provides:

- The definitional copular frame *What is X?* (where X is a nominalized concept);
- The processual frame *What is X-ing?* (e.g., "What is thinking?");
- The agentive frame *What is an X-er?* (e.g., "What is a thinker?");
- The adjectival frame *What does it mean to be X?* (e.g., "What does it mean to be conscious?");
- The infinitive frame *What is it to X?* (e.g., "What is it to love?").

All five frames are grammatically well-formed in English. Each, however, selects different distributional environments — different lexical neighborhoods in the embedding space, as Study 2 documents. The definitional copular frame, applied to invisible-process concepts, selects an entity-space neighborhood populated by abstract nouns referring to essences, definitions, and fundamental natures. The other four frames select neighborhoods populated by descriptions of processes, agents, qualities, and actions.

The point is not that grammar prevents the use of the four alternative frames. The point is that grammar makes the definitional copular frame **available** as one option among several. Without this availability, the institutional selection process documented in the next subsection could not operate.

This affordance perspective has an important consequence for cross-linguistic comparison. Languages differ not only in the frames they provide but in the **default** frame their grammar makes salient. As Study 1 shows, the definitional copular frame is far more prominent in the corpora of high-constraint languages than in those of low-constraint languages. This does not mean that low-constraint languages cannot formulate the frame — they can — but that the frame is one option among many with comparable structural salience. In high-constraint languages, the frame's structural prominence may make it the path of least cognitive and rhetorical resistance.

6.1.3 Institutional selection: definition as objective function

The second layer is **institutional selection**. Among the affordances grammar provides, philosophical institutions have historically selected the bare-noun copular form for a recognizable reason: it serves the institutional objective of producing definitions. This claim warrants careful articulation.

Philosophy, throughout much of its institutional history, has sought *definitions* — stable, context-independent characterizations of concepts that can be cited, contested, refined, and inherited across generations of practitioners. The institutional practices of philosophy — the seminar, the published treatise, the dialectical exchange, the canonical citation — are organized around the production and contestation of definitions. A claim of the form "X is Y" can be cited, contested, refined, and inherited. A claim of the form "When people X, what happens varies according to…" is harder to cite, harder to contest, harder to inherit, because it does not collapse into a single stable proposition.

The definitional copular frame is structurally well-suited to this objective. It produces a single nominal target, which can be defined in a single nominal predicate. The proposition "X is Y" is portable, citable, and inheritable. The proposition "Loving is the kind of unfolding mutual process that varies according to…" is less portable.

This functional fit between the definitional frame and the institutional objective generates selection pressure. Over generations of philosophical practice, formulations that fit the frame are more readily incorporated into the canon than formulations that resist it. The canon, in turn, reinforces the frame: students learn philosophy by reading questions in the form "What is X?" and answering them in the form "X is Y." The frame becomes the default not because grammar forces it, but because the institution selects for it.

This account is structurally a *selection-by-objective-function* model: a population of grammatical affordances exists in the language; an institutional process operates on that population with a particular objective (here, the production of definitions); affordances that fit the objective are preferentially retained and propagated; affordances that do not are correspondingly demoted. Other objectives — description, narration, instruction, exhortation — would select different forms from the same population. The institutional objective function is what determines which grammatical affordance is foregrounded. This model is consistent with general accounts of cultural transmission in which the unit of selection is not the bearer of a trait but the practice or norm that selects among trait-variants (Sperber, 1996; Henrich, 2015), applied here to the practices and norms of philosophical inquiry.

6.1.4 Social stabilization: shared discourse and the unification of subjects

The third layer is **social stabilization**. Philosophical practice is not solitary. It occurs in communities — seminars, conferences, journals, textbooks — that depend on shared question formulations. The definitional copular frame "What is X?" is particularly well-suited to communal discourse, for a specific structural reason: it **unifies the question-subject** into a single nominal target.

Compare two formulations of a question about love:

(1) "What is love?"
(2) "When two people enter into a sustained mutual relationship marked by care, attention, vulnerability, and the willingness to be transformed by the other, what features of that relationship, viewed from which perspective, deserve our attention?"

Formulation (1) has a single subject (*love*) and a single predicate position to be filled. It can be inscribed on a blackboard, referenced across a seminar, cited in a paper, and disputed across centuries of literature. Formulation (2) has multiple participants, multiple perspectives, multiple features — each of which might be foregrounded or backgrounded by different interlocutors. The communal coordination problem is harder.

The social pressure toward (1) is not a function of laziness or intellectual deficiency. It is a function of the structural requirements of communal discourse. A question that cannot be inscribed in a portable form cannot be sustained across the institutional structures (citations, courses, syllabi, encyclopedias) through which philosophy reproduces itself. Formulation (1) is reproducible; formulation (2) is not.

This social pressure interacts with the institutional selection pressure of §6.1.3. Together, they produce a stable equilibrium in which the definitional copular frame is the default form of philosophical inquiry — not because no alternatives exist, but because the alternatives fail to meet the joint structural requirements of definition-production and communal coordination.

A consequence: when philosophy attempts to formulate questions in non-default forms, the reformulations are often metabolized back into the default. A philosopher who poses the question "What is it to love?" (an infinitive formulation) typically proceeds, within a few sentences, to begin treating "love" as the question's target. The grammatical form has shifted, but the institutional and social pressures pull the subsequent discourse back toward the bare-noun frame.

6.1.5 *Conceptual asymmetry: invisibility as enabling condition*

The fourth layer is **conceptual asymmetry**: the funnel operates only on concepts whose referents lack direct observational anchors. This is the empirical finding of Study 2 — visible-process concepts resist reification, while invisible-process concepts undergo a substantial shift toward entity-space upon nominalization.

This asymmetry has a structural explanation. When a concept's referent is observable, the observation provides a *deictic resolution*: a way of answering "What is X?" by pointing. "What is running?" can be answered by pointing to a running person and saying "that is running." The deictic resolution short-circuits the entity-search reasoning that the question form would otherwise license. The nominalized concept is grounded in the observable referent, and the bare-noun form does not drift into entity-space.

For invisible-process concepts, no deictic resolution is available. "What is consciousness?" cannot be answered by pointing. There is no observable referent to ground the noun. In the absence of observational grounding, the noun drifts into the distributional environment that the embedding analysis identifies as entity-space: a region populated by abstract nouns whose neighbors are *essence*, *nature*, *substance*, *definition*. The question form, combined with the absence of observational grounding, licenses entity-search reasoning as the natural response.

This conceptual asymmetry is the enabling condition for the funnel. Without it, the grammatical, institutional, and social pressures of the preceding subsections would have no traction. Even philosophical institutions cannot generate hard-problem-type questions about *running* or *eating*, because the observability of the referent prevents the noun from drifting into entity-space. The funnel requires both the institutional selection of the definitional frame and the conceptual invisibility of the target concept.

6.1.6 *The co-evolutionary loop*

The four layers — grammatical affordance, institutional selection, social stabilization, and conceptual asymmetry — do not operate independently. They reinforce one another in a co-evolutionary loop:

1. **Grammatical affordance** makes the definitional copular frame available as one option.
2. **Institutional selection**, driven by the objective function of definition-production, foregrounds this option over alternatives.
3. **Social stabilization** propagates the foregrounded option through the institutional structures of communal discourse, making it the default.
4. **Conceptual asymmetry** ensures that, when the default is applied to invisible-process concepts, the resulting questions stabilize as searches for essences.
5. The stabilized questions become the **canonical problems** of the tradition. New practitioners learn to formulate questions in the canonical form.
6. The canonical form **reinforces** the grammatical affordance's salience, the institutional objective, and the social default — closing the loop.

No single layer is sufficient. Grammar without institutional selection would leave the definitional frame as one option among many. Institutional selection without grammatical affordance would have no frame to select. Social stabilization without invisibility would propagate questions that are deictically resolvable (and therefore unproblematic). Invisibility without the institutional and social layers would produce occasional puzzlement but not stable traditions of philosophical inquiry.

The loop accounts for both the cross-linguistic distributional differences documented in Study 1 (high-constraint languages have more salient grammatical affordances, amplifying the loop's frequency) and the distributional-semantic differences documented in Study 2 (invisible concepts show larger embedding shifts because they lack deictic anchors).

6.1.7 Why this account survives the Sapir-Whorf objection

The standard objection to claims of linguistic relativity is that they over-attribute cognitive consequences to grammatical structure. The Eskimo snow vocabulary case, for instance, was originally cited as evidence that lexical richness causes perceptual discrimination, but subsequent analysis (Pullum, 1989; Martin, 1986) showed both that the lexical count was overstated and that the causal direction was reversed: environmental conditions drive vocabulary, which then accompanies — rather than causes — perceptual practice. Strong linguistic relativity claims tend to dissolve under scrutiny.

The co-evolutionary account is structurally different from the strong-relativity claim and is therefore not subject to the standard objection. Three features of the account warrant explicit mention:

First, the account does not assert direct causation from grammar to thought. It asserts that grammar provides affordances, among which institutions select. The causal work is done by the joint operation of all four layers, not by grammar alone. A reviewer who attempts to refute the account by showing that grammar alone is insufficient confirms, rather than refutes, the account: grammar alone is indeed insufficient; that is precisely the claim.

Second, the account predicts cross-linguistic variation in the *prevalence* of entity-search questions, not in their *availability*. All languages with copula constructions can formulate the definitional frame. The prediction is that languages with high copula constraint (Shared predication, required copula, high versatility) will exhibit greater corpus-level prominence of the frame and, by extension, greater institutional propensity to canonize the frame in philosophical traditions. Whether this prediction holds is an empirical question that the present study has partially addressed (corpus prominence is confirmed) and that subsequent work can address more fully (institutional propensity in historical philosophical corpora).

Third, the account makes specific predictions about which concepts undergo reification and which do not. The visibility asymmetry — invisible concepts shift to entity-space, visible concepts do not — is not a feature of strong linguistic relativity claims. It introduces a conceptual variable that interacts with the grammatical variable, generating a $2 \times n$ design (visibility $\times$ grammatical form) rather than a 1-dimensional claim about language effects on thought. This interaction structure is testable and falsifiable in ways that strong relativity claims typically are not.

The co-evolutionary account, therefore, is not a covert resurrection of strong linguistic relativity. It is a structurally distinct hypothesis about the interaction of grammatical, institutional, social, and conceptual factors in the production of philosophical questions.

6.1.8 Summary

The grammatical funnel does not operate through grammar alone. It operates through the joint and mutually reinforcing action of four layers: the grammatical affordance that makes the definitional copular frame available, the institutional selection that foregrounds this affordance over alternatives, the social stabilization that propagates the foregrounded form through communal discourse, and the conceptual asymmetry that ensures the form generates entity-search questions specifically for invisible-process concepts.

The present account is, in this sense, neither a strong claim about language determining thought nor a denial that language matters. It is a claim about how language and institution co-evolve in the production of theoretical questions — and about how, in the case of philosophical inquiry into invisible processes, the co-evolution has converged on a question form that generates hard problems as a structural consequence rather than as an inevitable feature of the phenomena themselves.

6.2 Why visible processes resist the funnel

Study 2 provides distributional evidence that visible-process concepts (*running*, *swimming*, *dancing*) remain strongly process-anchored across grammatical forms, while invisible-process concepts occupy comparatively more entity-oriented positions, especially in definitional copular frames. The relevant effect is therefore not a uniform absolute shift of invisible nouns into entity-space, but an interaction between grammatical form and the availability of observational anchoring. This asymmetry has a structural explanation grounded in **observational accessibility**.

When the referent of a concept is directly observable, the observation provides what we have termed a *deictic resolution*: "What is running?" can be answered by pointing — *"that is running"* — at a running person. The deictic resolution short-circuits the entity-search reasoning that the bare-noun copular form would otherwise license. The nominalized concept remains anchored to its observable referent, and the distributional environment of the noun reflects this anchoring.

For invisible-process concepts, no analogous deictic resolution is available. *Consciousness* cannot be pointed to. *Justice* cannot be exhibited. *Beauty* cannot be located. In the absence of observational grounding, the bare noun drifts into the distributional environment populated by abstract entity-terms: *essence*, *nature*, *substance*, *definition*. The question form then licenses entity-search reasoning as the structurally appropriate response.

This observational-anchoring account is consistent with prior findings on the cognitive role of demonstrative reference (Diessel, 2006) and on the special status of perceptually accessible concepts in linguistic processing (Connell & Lynott, 2014). It does not require any further theoretical machinery beyond what the embedding data already document.

6.3 Cross-linguistic comparison: a circumscribed prediction

The present findings predict that languages with low copula constraint — those classified as LOW in Section 3.2.4 — may afford structurally different question formats for theoretical inquiry. Japanese does not lack the bare-noun definitional frame — 「X とは何か」is grammatically available and used — but provides competing process-preserving frames with higher structural availability. For example, the construction 「X するとはどういうことか」(*X-suru to wa dōiu koto ka*, "what is it to do X?"), in which the verbal form *X-suru* is preserved and the question targets the activity rather than a nominal entity. Korean and Chinese, both classified as LOW-constraint, provide analogous constructions.

We do not, in the present paper, claim that the philosophical traditions of these languages have in fact developed differently from Anglophone or other high-constraint traditions. Such a claim would require a parallel corpus study of philosophical writing in these languages, and is beyond the present scope. The circumscribed prediction is structural: the grammatical affordances of low-constraint languages provide a wider range of non-reifying question forms than those of high-constraint languages, and the prevalence of such forms in canonical philosophical discourse is an empirical question deserving systematic investigation.

6.4 Limitations

Several limitations of the present study should be noted.

UD treebank genre composition. Universal Dependencies treebanks vary in genre — news, web text, fiction, parliamentary proceedings — and these genre differences may interact with the constructions we examine. The CopSubj rate, for example, is likely higher in expository registers than in narrative ones. We mitigated this by including treebanks with mixed-genre composition where available, but residual genre effects cannot be fully excluded.

WALS classification granularity. The WALS features 119A and 120A code each language with a single categorical value, abstracting over dialect, register, and historical variation. The Japanese coding as 119A-Split, for instance, reflects standard Japanese; informal speech may exhibit more flexibility. The categorical typology is a useful first approximation but does not capture intra-language variation.

Embedding model training bias. Both OpenAI text-embedding-3-large and gemini-embedding-001 are trained predominantly on English-language corpora. The cross-validation reported in Section 4.3.2 controls for model-specific architectural artifacts, but does not control for shared training-data bias. Future work should examine the embedding-space distributions of nominalized forms in models trained predominantly on non-English data.

Anchor phrase construction. The Study 2 design uses three independent anchor sets (essence vs. activity; object vs. event; definition vs. change) of five phrases each. The absolute magnitudes of E-P gaps vary substantially across anchor sets, but the interaction direction is consistent across both anchor formulation and embedding model (8/8 grammatical forms show invisible-class > visible-class in entity-space across both OpenAI and Gemini). The robustness of the relational structure is good; the absolute magnitudes of effects, however, should not be over-interpreted.

Visibility operationalization. The visible/invisible distinction central to the interaction effect documented in Section 4.3 was applied by the authors but is independently validated by Brysbaert et al.'s (2014) concreteness ratings (Section 4.4). Future work might extend the operationalization with modality-specific perceptual strength norms (Lynott & Connell, 2009) and might examine whether the funnel effect varies smoothly with concreteness rather than dichotomously.

Philosophical corpus. The claim that many canonical formulations in Western philosophy privilege the bare-noun copular frame is supported here only by example and citation (Section 1.1) rather than by a systematic corpus analysis of philosophical writing. A dedicated study of philosophical text corpora (e.g., the Stanford Encyclopedia of Philosophy, JSTOR philosophy journals, canonical primary texts in translation) would convert this claim from an illustrative observation to a documented finding.

Psychological verification. The findings reported here concern distributional and typological structure. Direct verification of whether human reasoners produce more entity-search responses under definitional copular frames than under processual alternatives remains for future work.

7. Conclusion

This paper has introduced "Thinking for Theorizing" — the claim that grammatical structure constrains not everyday cognition but the formal structure of theoretical inquiry — and provided converging evidence for the grammatical funnel hypothesis.

Three layers of evidence support the claim:

1. **Typological.** Languages with shared nominal-locational predication, required copula, and high copula versatility exhibit substantially higher rates of noun-subject copular constructions used as structural proxies for the definitional frame (HIGH/LOW ratio = 2.63×), independent of language family.

2. **Distributional-semantic.** Across two independent embedding models (OpenAI text-embedding-3-large and Google gemini-embedding-001) and three independent anchor formulations, invisible-process concepts occupy more entity-oriented — or less process-anchored — distributional positions than visible-process concepts in all eight grammatical forms tested (8/8 cross-model agreement; Pearson $r = 0.73$ between models). Visible concepts remain process-anchored even under nominal and copular formulations, while invisible concepts show the strongest relative entity-orientation in definitional copular frames (cross-anchor sign agreement = 83%).

3. **Theoretical.** English provides four classes of grammatical devices — processual, agentive, gradable adjectival, and infinitival — that preserve the processual character of a concept. Many canonical formulations in Western philosophy privilege the bare-noun copular form under an institutional objective of definition. The resulting entity-search reasoning is not a consequence of grammatical necessity but of co-evolutionary selection among available grammatical affordances.

The definitional copular frame "What is X?" is not the only way to pose fundamental questions. It is a choice — but a choice so deeply habituated in philosophical discourse that it appears to be the only way. The present study reveals the grammatical architecture of that habit.

8. Appendix A. Cross-linguistic corpus summary

Full data for the 15 Universal Dependencies treebanks analyzed in Study 1. Token counts are total tokens analyzed; rates are per 1,000 tokens. WALS classifications follow Stassen (2013a, 2013b). Copula head entropy is the Shannon entropy of the part-of-speech distribution of copula heads (`cop` dependency relation in UD).

Table A1. Cross-linguistic corpus summary.

Lang	Name	Family	Tokens	Cop rate	CopSubj rate	N/V ratio	Cop entropy	WALS 119A	WALS 120A
ar	Arabic	Afro-Asiatic	223,881	1.63	0.84	4.44	0.999	Split	Zero OK
de	German	Indo-European	263,777	13.33	6.46	2.61	1.092	Split	Required
en	English	Indo-European	204,578	23.11	7.61	1.54	1.201	Shared	Required
es	Spanish	Indo-European	382,327	12.86	4.21	2.18	0.979	Split	Required
fi	Finnish	Uralic	162,815	26.74	15.42	2.07	1.147	Shared	Required
fr	French	Indo-European	354,647	13.88	6.05	2.37	1.266	Split	Required
hi	Hindi	Indo-European	281,057	9.73	3.21	2.14	0.936	Shared	Required
id	Indonesian	Austronesian	97,602	8.76	2.45	2.13	0.419	Split	Zero OK
it	Italian	Indo-European	276,014	11.60	6.03	2.35	1.086	Split	Required
ja	Japanese	Japonic	168,333	12.49	2.48	2.76	0.000	Split	Required
ko	Korean	Koreanic	296,446	0.91	0.05	1.59	0.486	Split	Required
pt	Portuguese	Indo-European	171,776	11.84	4.88	1.99	1.272	Split	Required
ru	Russian	Indo-European	74,900	2.19	0.73	3.16	0.954	Shared	Zero OK
tr	Turkish	Turkic	100,713	24.39	7.49	1.79	1.180	Shared	Required
zh	Chinese	Sino-Tibetan	98,614	13.04	5.52	1.85	0.345	Split	Zero OK

Notes. Cop rate = total `cop` tokens per 1,000 tokens. CopSubj rate = `cop` tokens whose dependent (subject) is a noun, per 1,000 tokens. N/V ratio = noun token count divided by verb token count. Cop entropy = Shannon entropy over {Noun, Adjective, Verb} as the head of `cop`-dependent clauses; higher values indicate that the copula serves more diverse predication categories (the "master-key" property discussed in §3.2.2).

9. Appendix B. Embedding stimulus list

Full stimulus list for Study 2 (n = 126 phrases). Each concept × form combination produces 1–2 phrases. Forms with no available agentive for the concept (e.g., *consciousness*, *eating*) are omitted from the agentive row.

Table B1. Stimuli by concept and grammatical form.

Concept	Class	Verb	Progressive	Agentive	Noun (minimal)	Noun (+ word)	Noun (+ feeling/act)	*what is X*	*X is something*	*X means*
love	invisible	*to love; people loving*	*loving; loving something*	*a lover of music*	*love*	*the word love*	*the feeling of love*	*what is love*	*love is something*	*love means*
consciousness	invisible	*to be conscious of something; someone being conscious*	*being conscious; being conscious naturally*	—	*consciousness*	*the word consciousness*	*the feeling of consciousness*	*what is consciousness*	*consciousness is something*	*consciousness means*
truth	invisible	*to tell the truth; someone being truthful*	*being truthful; being truthful naturally*	—	*truth*	*the word truth*	*the feeling of truth*	*what is truth*	*truth is something*	*truth means*
beauty	invisible	*to be beautiful; someone being beautiful*	*being beautiful; being beautiful naturally*	—	*beauty*	*the word beauty*	*the feeling of beauty*	*what is beauty*	*beauty is something*	*beauty means*

Concept	Class	Verb	Progressive	Agentive	Noun (minimal)	Noun (+ word)	Noun (+ feeling/act)	*what is X*	*X is something*	*X means*
justice	invisible	*to act justly; someone acting justly*	*acting justly; acting justly naturally*	—	*justice*	*the word justice*	*the feeling of justice*	*what is justice*	*justice is something*	*justice means*
freedom	invisible	*to be free; someone being free*	*being free; being free naturally*	—	*freedom*	*the word freedom*	*the feeling of freedom*	*what is freedom*	*freedom is something*	*freedom means*
running	visible	*to run; someone running*	*running; running naturally*	*a runner*	*running*	*the word running*	*the act of running*	*what is running*	*running is something*	*running means*
swimming	visible	*to swim; someone swimming*	*swimming; swimming naturally*	*a swimmer*	*swimming*	*the word swimming*	*the act of swimming*	*what is swimming*	*swimming is something*	*swimming means*
dancing	visible	*to dance; someone dancing*	*dancing; dancing naturally*	*a dancer*	*dancing*	*the word dancing*	*the act of dancing*	*what is dancing*	*dancing is something*	*dancing means*
eating	visible	*to eat; someone eating*	*eating; eating naturally*	—	*eating*	*the word eating*	*the act of eating*	*what is eating*	*eating is something*	*eating means*
walking	visible	*to walk; someone walking*	*walking; walking naturally*	*a walker*	*walking*	*the word walking*	*the act of walking*	*what is walking*	*walking is something*	*walking means*

ConceptClass	Verb	Progressive	Agentive	Noun (minimal)	Noun (+ word)	Noun (+ feeling/act)	*what is X*	*X is something*	*X means*
cooking visible	*to cook; someone cooking*	*cooking; cooking naturally*	*a cook*	*cooking*	*the word cooking*	*the act of cooking*	*what is cooking*	*cooking is something*	*cooking means*

Note. "Noun (+ feeling/act)" is *the feeling of X* for invisible concepts and *the act of X* for visible concepts, reflecting the natural collocational expectations of each class. The agentive row uses "—" where no standard English agentive form exists.

10. Appendix C. Anchor pole specifications

Three independent anchor sets were used in Study 2 to compute the entity-process (E-P) gap. Each pole is the centroid of five phrases; the E-P gap for a stimulus is its cosine similarity to the entity centroid minus its cosine similarity to the process centroid.

Anchor Set 1 — Essence vs. Activity

Pole	Phrases
Entity	*an essence; a fixed nature; an eternal kind; the underlying being; what something fundamentally is*
Process	*an activity; an ongoing doing; something happening; people performing actions; what someone is doing*

Anchor Set 2 — Object vs. Event

Pole	Phrases
Entity	*an object; a thing that exists; an entity; a permanent item; a stable substance*
Process	*an event; something that occurs; an episode; an unfolding occurrence; an action in time*

Anchor Set 3 — Definition vs. Change

Pole	Phrases
Entity	*something that has a definition; a category with criteria; a concept with fixed meaning; an item that can be characterized; a referent that can be specified*
Process	*an ongoing change; a transformation; something developing; a continuous flow; movement through time*

Notes. The three anchor sets were designed to capture conceptually distinct facets of the entity/process distinction: ontological depth (essence/activity), permanence (object/event), and characterizability (definition/change). The cross-anchor sensitivity analysis in §4.5.2 examines whether the visibility × form interaction is preserved across these independent operationalizations.

11. Appendix D. Brysbaert concreteness ratings for study concepts

Concreteness ratings from Brysbaert, Warriner, and Kuperman (2014) for the concept forms used in Study 2. Ratings are on a 1–5 scale (1 = most abstract, 5 = most concrete), each averaged from approximately 30 naive raters. The visible/invisible classification is fully separated on the concreteness dimension (Cohen's $d = 5.08$).

Table D1. Brysbaert concreteness ratings.

Class	Word	Concreteness
Invisible	*lover*	3.68
Invisible	*beauty*	2.93
Invisible	*freedom*	2.34
Invisible	*consciousness*	2.32
Invisible	*beautiful*	2.16
Invisible	*love*	2.07
Invisible	*free*	2.04
Invisible	*conscious*	2.03
Invisible	*truth*	1.96
Invisible	*truthful*	1.90
Invisible	*loving*	1.73
Invisible	*just*	1.52
Invisible	*justice*	1.45
Visible	*runner*	4.86
Visible	*swimmer*	4.77
Visible	*dancer*	4.75
Visible	*swimming*	4.44
Visible	*eat*	4.44
Visible	*swim*	4.43
Visible	*walker*	4.42

Class	Word	Concreteness
Visible	*dance*	4.32
Visible	*cook*	4.32
Visible	*run*	4.31
Visible	*dancing*	4.30
Visible	*walking*	4.29
Visible	*running*	4.27
Visible	*cooker*	4.22
Visible	*cooking*	4.19
Visible	*eating*	4.14
Visible	*walk*	4.07
Visible	*eater*	3.75

Notes. Mean concreteness for invisible-class words: 2.16 (range 1.45–3.68). Mean for visible-class words: 4.35 (range 3.75–4.86). Within-class outliers (*lover* at 3.68 on the invisible side; *eater* at 3.75 on the visible side) are nearly the only points where the two distributions approach each other, and the gap between them is below 0.1 on a 5-point scale.

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End of draft v1. Target: Cognitive Linguistics / Language and Cognition.